Teddy Liu

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EDUCATION

University of California, Davis Davis, CA
Bachelor of Science, Aerospace Engineering *Expected December 2027*
Bachelor of Science, Mechanical Engineering

EXPERIENCE

UC Davis Memorial Union Games Area Davis, CA
Student Mechanic *November 2023 – Present*

- Maintain, troubleshoot, and repair high-cycle industrial machinery, including electromechanical pinspotters and automated lane conditioning platforms.
- Diagnose mechanical drive assemblies, gearboxes, microswitches, and fluid delivery pumping systems using systematic fault-isolation techniques to minimize lane downtime.
- Inspect and service mechanical power transmission components, including belt drives, solenoid actuators, and bearings, ensuring continuous operational uptime across all lanes.
- Coordinate maintenance scheduling and machine status updates across the team, tracking repairs and flagging recurring electro-mechanical issues in real time.

Alameda County Public Works Agency Hayward, CA
Engineering Intern *July 2022 - August 2022*

- Utilized CAD software to catalog and map pedestrian ramp infrastructure across county road networks in support of ADA compliance surveying.
- Contributed to infrastructure documentation, providing accurate spatial data to support road improvement planning and civil engineering operations at the county level.

RELEVANT PROJECTS

DIY Equatorial GoTo Telescope Mount Tracker April 2026 – Present

- Engineered an automated GoTo equatorial tracking mount for refractor and reflector telescopes, running OnStepX open-source firmware on an ESP32 microcontroller with TMC2209 silent stepper drivers.
- Designed and fabricated a GT2 belt reduction drivetrain for the Right Ascension (RA) axis, calculating the optimal reduction ratio to balance output torque with high-resolution sidereal tracking accuracy.
- 3D-modeled custom optical tube adapters in Autodesk Fusion 360, printed in Matte PLA with a high-perimeter gyroid infill pattern to maximize structural rigidity and eliminate flexure under dynamic payloads.

Optoisolated Network Wake Switch *December 2025*

- Developed a remote power management system using an ESP32 microcontroller to interface with a PC motherboard's front-panel power headers via PC817 optocouplers, ensuring galvanic isolation.
- Engineered a custom snap-fit enclosure in Fusion 360 utilizing precise dimensional tolerances to house microcontrollers, wiring terminals, and status LEDs, fabricated on an FDM 3D printer.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks, Autodesk Fusion 360, Onshape, CAD Modeling, GD&T Fundamentals, Kinematic Analysis, Design for Manufacturing (DFM), Structural Optimization, Statics & Dynamics, Thermofluids.

Mechatronics & Hardware: Embedded Systems (ESP32), Stepper Motor Control, TMC2209 Drivers, Actuators & Drivetrains, Optoisolators, Automotive Electrical Integration, Soldering & Circuit Diagnostics.

Fabrication & Maintenance: FDM 3D Printing (PrusaSlicer, Parameter/Infill Optimization), Rapid Prototyping, Mechanical Assembly, Electromechanical Machine Maintenance, Fluid Delivery Systems.

Software & Tools: Python, C/C++, Bash, Linux (Ubuntu/Debian CLI & Headless), Docker, Git, Networking & Reverse Proxies.